

Machine Learning-Based Safety System for Women with Real- Time Alerts and Geo-Tracking

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ABSTRACT Increasing mobility, dynamic social interactions, and delayed emergency response mechanisms have made it extremely difficult to ensure women's safety in quickly growing urban environments. Traditional safety systems are less effective in high-risk and real-time scenarios because they frequently rely on manual activation or preset rules. The development of intelligent, automated safety monitoring solutions is made possible by recent developments in machine learning techniques and wearable and smartphone-embedded sensor technologies. This study offers a machine learning-based framework for women's safety that uses ongoing motion sensor data analysis to spot unusual and possibly dangerous human activity patterns. The suggested system uses gyroscope and accelerometer data from smartphone sensors to identify abnormalities and identify human activity. With regard to conducting an experimental analysis, a commonly accepted UCI Human Activity Recognition (HAR) dataset is used for reliable and consistent outcome analysis. The sensor data tends to undergo a series of preprocessing techniques like noise reduction, normalization, and feature extraction for enhanced classification outcomes. Three individual supervised learning techniques like Support Vector Machines (SVM), K-Nearest Neighbor (KNN), and Random Forests (RF) are implemented with regard to classification outcomes for human activity patterns. Furthermore, a soft voting ensemble model is developed by merging all individual classifiers for more reliable decision-making outcomes. The proposed framework is analyzed based on appropriate classification performance using parameters like accuracy, precision, recall, and F1 score. The experimental outcomes show that the ensemble learning method significantly outperforms each of the classifiers on their own, with an overall accuracy of 95 percent for recognizing abnormal/risky activity patterns. This shows the efficiency of ensemble learning methods for handling noisy sensor readings, reducing misinformation, and improving the overall efficiency of the system when dealing with real-time data. Also, it should be noted that the system utilizes geo-tracking features as well as alert mechanisms to reduce response times when dealing with emergency situations, thus providing a more effective platform for real-time sharing of locations and instant notifications of emergency contacts when abnormal activities are recognized. From an overall perspective, it can be said that the presented machine learning-based safety system has shown promising efficiency as a greatly adaptive system for real-time women's safety monitoring with the integration of effective intelligent activity recognition, ensemble learning, and alert generation functions.

I. INTRODUCTION

The fusion of machine learning (ML) with wearable sensor technology has profoundly changed the manner in which human activities, behavior, and safety conditions are dynamically monitored and assessed. Wearable sensors, which are embedded with accelerometers, gyroscopes, and position sensors, provide real-time information that enables intelligent machines to detect human activities and abnormal behavioral patterns with high accuracy. However, it has been observed that traditional security and personal protective mechanisms, especially those developed for women's protection, have shown limited effectiveness, as they are generally dependent on various activities such as manual intervention, predefined decision rules, and fixed operation modes. These traditional security measures are observed to be ineffective, especially during emergency situations, as they rely on heavy manual intervention methods, such as decision-making through alert messages, leading to reduced effectiveness. Moreover, it has been observed that existing personal protective mechanisms are generally developed with fixed assumptions, leading to reduced adaptability under real-time situations where there is a high probability of changing human behaviors. Despite the extensive research that exists on ML-based predictive and monitoring systems across different domains, like healthcare, education, and public safety, a majority of the existing ML-based systems primarily focus on utilizing offline or past data for making decisions. Nevertheless, the focus on past experimental results in most ML systems for safety is a cause of concern, given the difficulties that arise when interpreting behavioral changes or emerging patterns. Second, the majority of the existing ML-based safety solutions have been utilizing the traditional approach of a single-model learning approach, where a lot of emphasis is given to maintaining the stability of the machine learning model and the robustness of the results. However, the existence of a lot of complex and unreliable sensor data is not properly addressed. Hence, the proposed solution outlines the underlying requirements for the implementation of a machine learning approach that incorporates wearable sensor technology for

continuous learning. The system can process real-time motion patterns, GPS-based location data, as well as contextual information related to environmental factors like weather conditions to identify abnormal or hazardous situations that may arise. Also, by implementing the ensemble learning technique, the proposed model takes advantage of various machine learning classifiers, which increases the overall reliability of the decision made by the model despite errors that may come about as a result of individual classifiers. Further, the system can recognize abnormal patterns of behavior like sudden falls, lack of activity, or unusual patterns of movement, which may present an accident, hazardous, or unhealthy situation. Once it identifies abnormal patterns, it sends out automatic notifications as well as the real-time location of the person to predetermined family or emergency contacts without the need to perform any manual action by the individual, thus enabling the efficiency of the personal safety device. Furthermore, the adaptive nature of ensemble learning allows the proposed framework to remain performing well even in the presence of sensor noise, missing data, or environmental variability. By continuously learning from incoming streams of data and contextual information, the system can adapt to evolving user behaviors and operational conditions, assuring consistent accuracy in monitoring. Overall, the proposed configuration embodies a context-aware and resilient solution that overcomes the limitations of traditional safety systems by integrating real-time sensor intelligence, adaptive machine learning, and automated alert generation and provides a robust, scalable approach for enhancing personal safety and the effectiveness of emergency responses.

II. LITERATURE REVIEW

The advancements achieved in the field of pervasive computing, wearable technologies, IoT, and machine learning have added significant value to the design of intelligent monitoring systems and safety tools. These tools are primarily developed with the capabilities of detecting and analyzing the activities of humans and raising alerts accordingly. There has been significant advancement achieved in the field of continuous

monitoring solutions through the integration of intelligent analytics and wireless communication technologies to various domains such as the safety of women, healthcare systems, and security systems. Strong literature has been developed with focus on the design of activity recognition tools with the added feature of alert systems. [1] introduced a sensor-based machine learning framework that enables automatic recognition of human physical activities and detection of possible potential threats based on human motion. In their research, wearable sensors are used for collecting data from accelerometers and gyroscopes or gyroscope sensors. The collected data are then passed through machine learning algorithms to recognize normal and abnormal activities. This research explained how automatic detection of human activities based on motion can detect distress patterns or dangerous situations. In another research article, Kabir and Tasneem introduced a smart wristband system that uses a mobile application to increase situational awareness among people. This system identified emergency situations by considering contextual as well as motion-based patterns. After identifying such situations, this system automatically sends emergency notifications to the contact list. The research work carried out by Ganesan et al. on "IoT-based Health Monitoring Model for Prediction of Cardiac Problems" has confirmed the applicability of integrating ML with IoT technology. The viability of the integration technology of IoT/ML for effective problem prediction has been established. A research work on "Human Activity Recognition Using Smart Sensors" has been reported by Mehra et al. The significance of using effective techniques for efficient feature extraction using a human activity recognition system was emphasized. A comparative study was carried out by the author to understand the impact of different features on the efficiency of the human activity recognition system. It has been reported that the system achieves improved performance with effective feature engineering. Akram et al. proposed an IoT-based safety system. A wearable IoT-based safety device has been developed by the author. It has been reported that the device enables effective real-time monitoring. It alerts in case of emergency. In the proposed study, accelerometer,

GSM, and GPS are used for monitoring human movement using a wearable IoT-based safety device.

Sunehra et al. proposed a raspberry pi-based wearable safety system that also included GPS and machine learning algorithms capable of recognizing data and abnormalities within the data that could reflect problem situations. After recognizing an abnormality via machine learning algorithms, an alert was sent to the concerned authorities. The practicality of machine learning algorithms for activity recognition was proved in IoT devices with the proposed system. Shobitha et al. proposed an Arduino framework-based wearable device that was developed with the objective of improving the safety of women. The proposed system was developed with the aim of sending emergency messages along with location information during problem situations. The system was focused on the basic requirements of simplicity, cost-effectiveness, and reliability; hence, it was highly appropriate for implementation. Arul Gandhi et al. developed a smart wearable system based on gyroscopes and accelerometers that recognized physical activities using sensed data. The sensed data was used to identify physical activities such as walking, running, and movements. The study demonstrated the effectiveness of motion sensors in activity classification and laid groundwork for behavior-based safety applications. Rohini and Sangeetha [9] proposed a wearable system combining IoT and ML to monitor user movements and generate alerts using GPS and SMS services. The system emphasized continuous monitoring and automated alert generation without user intervention. The research proved the efficiency of motion sensors in activity classification and established the foundation in behavior-based safety applications. Rohini and Sangeetha [9] presented a wearable system that combined IoT and ML to track user movements and raise alerts using GPS and SMS services. The system underscored continuous monitoring with automatic generation of alerts devoid of user intervention. In this process, the research underscored the need for autonomy and real-time response in safety systems. Rani and Venkatesh [10] proposed an alert mechanism in real time using machine learning algorithms

combined with GPS tracking. The system identified abnormal movement patterns and communicated emergency information immediately. The study reinforced the efficiency of ML-driven decision-making for rapid threat detection.

Naik et al. [11] suggested a hybrid machine learning model for continuous health monitoring using automated alerts. This manuscript focused on robust and adaptable monitoring systems. In the manuscript, the authors proposed using multiple models of machine learning for improving the predictions' results' reliability and accuracy. Jadhav et al. [12] elaborated on the importance of using contextual information data in improving the reliability of safety prediction systems. The manuscript focuses on the results of evaluating the application of multiple models of machine learning and demonstrated the significance of using multiple-dimensional data analysis. The results of the manuscript demonstrated the importance of using contextual information together with sensor data for improving the accuracy of the results. Naidu et al. [13] proposed an alarm system with the ability to generate alerts using machine learning predictions of vulnerable zones. This research showed that spatial intelligence could be utilized to improve safety decision making. Singh et al. [14] presented an edge computing-supported response system that utilized edge computing to process data locally to minimize latency during responses. Latency was minimized by reducing dependence on cloud infrastructure. Further, this research emphasized the significance of latency in safety responses. Jagadeesh et al. [15] presented an alert system that utilized cloud infrastructure and Firebase for providing alerts in monitoring GPS and temperature data. This research showed the capability of cloud infrastructure in managing large amounts of data.

Sethi & Sarangi [16] offered a comprehensive overview of the various architectures, protocols, and applications of the Internet of Things. Their study presented the IoT as a layered structure, i.e., perception, transport, processing, middleware, and application layers. Lightweight protocols such as MQTT and CoAP were emphasized. This study

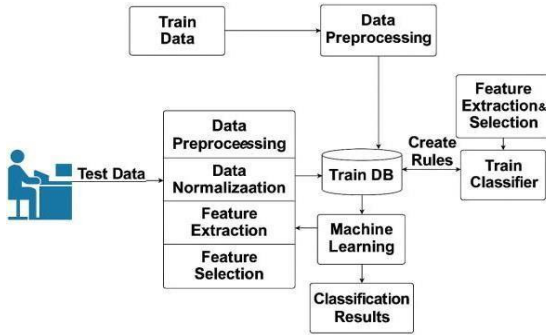
offers a basic IoT-based architectural perspective of sensor-based monitoring systems. Lara & Labrador [17] offered an exhaustive survey on human activity recognition systems based on wearable sensors. Their study focused on various sensor modalities, data preprocessing, feature extraction, and machine-learning-based classification. Several challenges were identified: noise emanating from the sensors, variability of behavior among the human subjects, as well as energy efficiency. Aggarwal & Ryoo [18] offered an exhaustive review of various techniques used for human activity analysis. Several conventional as well as learning-based machine intelligence-based methods were discussed. Banos et al. [19] researched human activity recognition by benchmarking many data mining and machine learning methods. Their findings provide a guideline on how to choose the most appropriate machine learning method through a comparative study of the results from various machine learning approaches. Chen et al. [20] researched various aspects of activity recognition based on sensor systems with an emphasis on context-aware/adaptive systems for the recognition of human activities. From the various research articles on the topic, it is evident that there is a significant amount of research interest in the integration of IoT systems with wearable sensor technology based on machine learning paradigms for the development of intelligent activity recognition systems. Although the intelligent systems reviewed are reliable, there are still many challenges with the systems, such as the variability of the input data, energy efficiency, real-time results, privacy, etc.

III. METHODOLOGY

The proposed system can operate on top of an existing system which is a machine learning pipeline. This pipeline must have the ability to operate in real-time to act as an activity detector/emergency notifier. The entire process can be considered a multi-step procedure consisting of data collection, data preprocessing, feature extraction, model training using the extracted feature set along with test data, evaluating, and then running a simulation in real-time to test the completeness of speed and

accuracy.

Fig. 1 Methodology of ML based system



The source of the data for the experiment is the UCI Human Activity Recognition (HAR) dataset. The UCI HAR dataset has tri-axial data from the accelerometer and gyroscope of the individuals from the smartphone, which was used to collect the data from the various projects regarding the activities of the individuals with reference to the 6 physical activities, such as walking, walking upstairs downstairs, standing, lying minute, among others. These activities were represented by a time value, and each was linked to one of the 6 activities, making them appropriate for the controlled activity low rate of the miners, which is quite dependent a priori on linking the various activities with reference to the data from the sensors. The construction of the process was intended to establish the vital features of the data from the sensors. All time : mean, standard deviation, energy, skewness, kurtosis, and entropy using the raw signals. All frequency : which leads to the use of key frequencies, FFT coefficients, and spherical entropy.

When it comes to features, this system maximizes the performance ability to make:

Random Forest Importance: This method was used to rank the features using the Gini impurity decrease technique. Recursive Feature Elimination (RFE): This method was used to eliminate the low-ranking variables to improve performance. At the data preprocessing stage, it was ensured that the data was fully trainable and directly usable:

The missing values were handled for the data as follows: Replace missing values with interpolation and using the forward fill method.

Noise filtering: A Butterworth low-pass filter was used, which removed any unwanted oscillations that the sensor was detecting and smoothed out the signal accordingly.

Normalization: Normalization of the values of the features was carried out by applying a Standard Scaler that gave balanced input for the development of the classifier.

IV. MACHINE LEARNING MODELS USED

The final outcome is achieved by using a supervised learning technique called Support Vector Machine (SVM). SVM works in the following manner: Each activity is classified into different classes using an optimal hyperplane that achieves the maximum margin between classes. This quality of SVM makes it extremely effective for use in complex and high-dimensional sensory spaces. In this proposed system, the Radial Basis Function (RBF) kernel mapping is used. This kernel is used because it is capable of addressing non-linear relationships that exist between class labels and input features. This ability of the RBF kernel also helps in the effective discrimination between overlapping activity patterns that are commonly found in human motion activity detection. SVM is an effective classifier to be used for activity recognition because it avoids overfitting. In addition, SVM is capable of generalization with fewer training examples.

The decision function of SVM is given by:

$$f(x) = \text{sign} \left(\sum_{i=1}^n \alpha_i y_i K(x_i, x) + b \right) \quad (1)$$

Where, X_i represents the support vectors, Y_i denotes class labels, $K(X_i, x)$ is the RBF kernel, b is the bias term

The margin-maximizing aspect of SVM proves to be useful while achieving an effective separation

of normal and abnormal activity patterns, which is very much required for safety-based applications.

K-Nearest Neighbor (KNN) is a non-parametric instance-based learner that classifies the test sample based on the class to which the majority of the k nearest neighbors belong in the feature space. Unlike other classifiers that follow a modeling-based approach to classification, the KNN algorithm is non-parametric; this means that the KNN algorithm doesn't follow the training phase model to arrive at the result. Instead, the entire dataset is stored during the computation phase:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \quad (2)$$

The KNN algorithm is easy to understand and works well because human activities are commonly similar and produce similar patterns. However, KNN is affected by noisy data. Therefore, normalization of features is used for preprocessing. This normalizes any given feature so that no feature dominates the others. Although KNN can be computationally expensive for large datasets, it works well as a baseline classifier.

Random Forest is an ensemble learning technique that uses a set of Decision Trees to make predictions. It trains a number of Decision Trees on the given dataset with randomly sampled sets of features. The predictions are made by aggregating all the outputs of the Decision Trees using majority voting. This ensures that all Decision Trees are different and overfitting is avoided. This is especially useful when the dimension of the data collected by sensors is high, as random forests handle noisy measurements with ease. The final prediction is:

$$\hat{y} = \text{mode}\{\mathcal{Y} - \{T_1(x), T_2(x), \dots, T_n(x)\}\}. \quad (3)$$

Where, $T_n(x)$ represents the predictions made by each decision tree. On one hand, Random Forest

is good when handling non-linear relationships as well as the interaction of variables without necessarily having to perform feature scaling. On another note, it can handle predictions even when there are sensor noises, which makes Random Forest very suitable when creating safety monitoring systems. Random Forest helps in feature importance, which can be analyzed to understand the aspect of attributes, which are significant in classifying activities, in sensor attributes.

The Voting Ensemble classifier utilizes the combined result of predictions using SVM, KNN, and Random Forest, resulting in better robustness and classification accuracy. Unlike using a single classifier, this voting ensemble utilizes a combination of models' strengths while compensating for their weaknesses, using a technique with soft voting, "where all predicted probabilities are summed and then divided by the number of models that provided a prediction":

$$P(y) = \sum_{i=1}^N P_i(y). \quad (4)$$

Instead, by integrating probabilistic outputs rather than hard class labels, the prediction produced by the ensemble model is smoother and more reliable. This approach is particularly advantageous in safety-critical systems, as it reduces misclassification of closely related activities such as walking upstairs and walking downstairs. The Voting Ensemble consistently outperformed the performance of single models by improving the generalization capability and reducing the minimum errors of classification; thus, it is considered the most reliable for real-time applications in women's safety.

V. CLASSIFICATION

In order to determine and implement the four classifiers: Random Forest, which is a combination of decisions trees that function well in higher dimensions and are robust to changes in

the input. A grid search is made on the parameters of the hyperparameters of the classifier, specifically for the parameters related to the number of estimators and the depth of the decision trees. SVM is a classifier that is based on a margin and uses the radial basis function. The system focuses on the tuning of the parameters, specifically for C and gamma, to improve the classification. In addition, a comparison classifier. KNN classifier is a non-parametric classifier that is created by measuring the euclidean norms.

Voting Ensemble: Voting ensemble is an ensemble method that involves RF, SVM, and KNN classifiers using soft voting as the principle of the ensemble method. In the current investigation, Voting Ensemble combines the probabilistic results of the individual classifiers in an effort to maximize the results and limit the error.

VI. RESULTS AND DISCUSSION

The analysis, training and visualisation components of the implementation relied upon several well known and generally used major libraries, namely. Scikit-learn, NumPy, pandas and matplotlib. The implementation of the algorithm was designed and run in a modular way

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

$$Precision = \frac{TP}{TP + P} \quad (6)$$

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

$$F1 \text{ Score} = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (8)$$

The

which allowed for the integration of various steps of the data processing and model evaluation sequences. The four models, Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbors (KNN) and Voting Ensemble were tested with conventional statistics, accuracy, precision, recall and f1 score. The results were shown in both tabular and graphical formats for easy assessment. Among all models, the Voting Ensemble was the most stable and consistent performer. It achieved an overall accuracy of 95 percent; the precision and recall values were above 0.95 for almost all classes of activity. Moreover, for "Laying," the score went perfectly to 1.00. That is very reliable performance against the stationary movement identification.

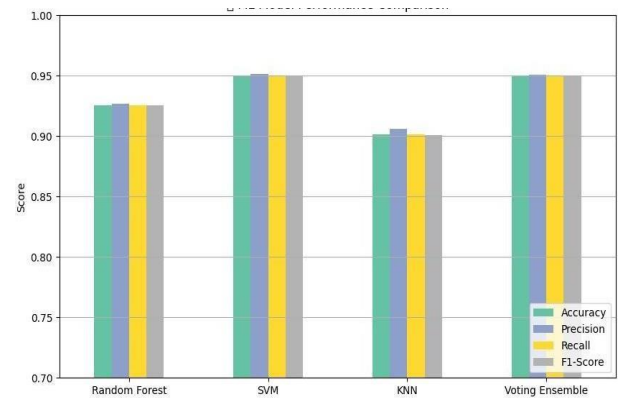


Fig. 2 Accuracy, precision, recall, and F1-score comparison of evaluated classifiers.

confusion matrix analysis was carried out to investigate the performance of the different models on different activity types. It is observed that the ensemble model results in a much lesser number of confusions between closely related activities, such as walking upstairs and walking downstairs-the critical requirement in safety-oriented systems where accuracy is reliability. In turn, Random Forest shows minor overlap between sitting and standing classes, which means occasional misclassification of the stationary postures. The SVM classifier has stronger class separation but with slight difficulties in transitional activities' separation. KNN showed higher misclassification rates due to its sensitivity to feature scaling and calculations of distances. This is mitigated by the Voting Ensemble because

it combined the advantages of these individual models by improving the consistency in classification and reducing the overall error.

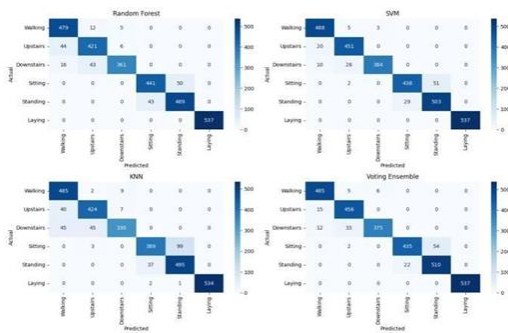


Fig. 3 Confusion matrices of Random Forest, SVM, KNN, and Voting Ensemble models.

Accuracy, precision, recall, and F1-score were some of the measures used in the proposed system. Random Forest and SVM exhibited higher accuracy on an individual basis than KNN. The overall best performance was given by the Voting Ensemble model, which achieved an overall accuracy of 95%, improving on the reliability and minimizing the misclassification between similar activities such as walking upstairs or downstairs. This thus verifies that indeed, ensemble learning enhances prediction stability for real-time women safety applications.

In order to test for responsiveness, a simulation was conducted with live data. The ensemble model classified inputs quickly without noticeable delay. In the event of detecting some sort of risk or abnormal condition-sudden immobility or prolonged resting, for instance-the system would send an alert to the user along with GPS information to the emergency contact. This was achieved using the Twilio API for SMS alerting, OpenStreetMap for a map of the real-time location, and all alarms were logged complete with timestamps, predicted activity, and alarm state: safe/alarm/emergency. This experiment showed the feasibility of the system in real-world applications on mobile and wearable platforms.

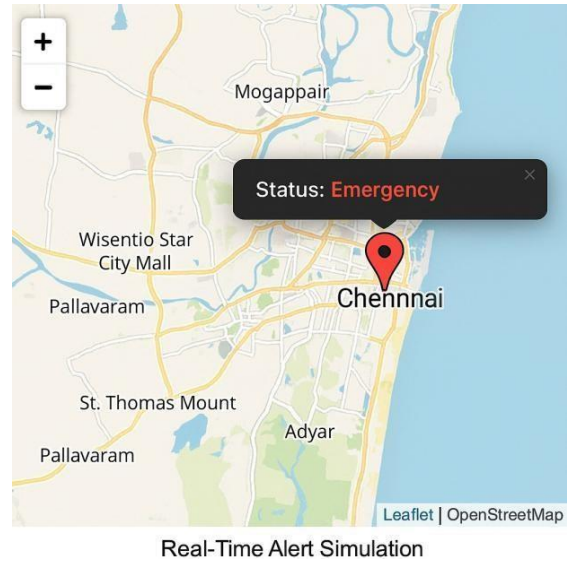


Fig. 4 Real-time emergency alert and GPS location visualization.

This figure demonstrates the simulation of the real-time alerts scenario in the proposed women safety system, in which an emergency condition is sensed by considering abnormal patterns of activity that are identified through the ensemble machine learning approach. The emergency condition is sensed, and at the same time, the current location of the user is shown on the digital map using OpenStreetMap services. The location marker indicates the exact location of the user at which the emergency situation is sensed, thereby providing effective situational awareness in emergency scenarios. This figure provides an idea about the applicability of the proposed machine learning-based women safety system in real-world scenarios, considering the ability of the system to sense the emergency condition in real time while providing location-based alerts that are essential for effective user safety.

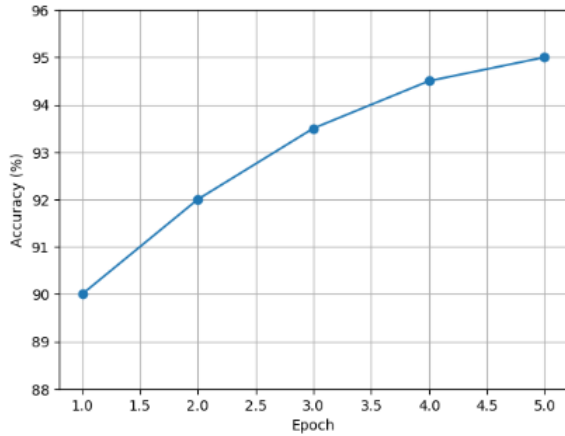


Fig. 5 Accuracy trend across Epoch

From the graph, we can see the variation of model accuracy based on the number of epochs used in the training process. It can be observed that as the number of epochs increases from 1 to 5, the model accuracy increases steadily from 90% to 95%.

VII. CONCLUSION

This project proposes a state-of-the-art machine learning-based safety system for women that incorporates real-time activity recognition and emergency alerts. The Voting Ensemble classifier showed superior performance compared to all individual classifiers by achieving high accuracy on all validation metric results, including accuracy, precision, recall, and F1. The application of GPS-based location tracking and emergency message sending greatly augments the applicability of the system by providing immediate assistance at times of emergencies. In this system, modularity and scalability were kept in mind to ensure that this system can easily be applied to mobile devices and wearable technologies with low latency and efficient computation. Real-time responsiveness of this system to emergencies was analyzed using simulated scenarios, and accuracy was checked using regular accuracy metric tests and feature analysis using a confusion matrix. Although this system exhibits significant potential and a sound methodology, it depends upon pre-labeled data

and may require data reset should this system be applied differently. Also, GPS accuracy depends upon device and external environment capabilities. Additionally, feature such as voice-based emergency detection is still in development. Further versions can comprise the inclusion of additional IoT devices, utilizing natural language processing algorithms that can be applicable to voice alerts, and acquiring real-time data instead of simulated data, so the system can be tested on a broader scope of field data. On a broader scope, it can be observed that the results achieved by the authors in the study have clearly established that the integration of ensemble learning, human activity recognition using sensor-based detection, and geo-tracking can help create a reliable solution for women safety applications. The proposed system clearly offers a proper ratio of accuracy, efficiency, and real-time execution, which plays an extremely significant role in safety-critical environments. The system minimizes errors during the detection of similar activities, thus creating an enhanced sense of trust. The study carried out by the authors has effectively established that safety systems powered by machine learning have the potential to become a fully integrated safety protection solution.

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